

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

09

10 answers · Rationale-rich · Teach from doc

Q1**D****D.** 4, 1

Choice D is correct. The solution to the system corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the quadratic equation shown intersect at the point $(4, 1)$. Therefore, $(4, 1)$ is the solution to this system.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2**A****A.** $5xx - 10y^2$

Choice A is correct. Since each term of the given expression has a factor of $5x$, it can be rewritten as $5xx - 5x10y^2$, or $5xx - 10y^2$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**B****B.** 0, -5

Choice B is correct. The y-intercept of a graph in the xy -plane is the point x, y on the graph where $x = 0$. At $x = 0$, the corresponding value of y is -5. Therefore, the y-intercept of the graph shown is 0, -5.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the y-intercept of a graph in the xy -plane that intersects the y -axis at $y = -4$, not $y = -5$.

Choice D is incorrect. This is the y-intercept of a graph in the xy -plane that intersects the y -axis at $y = 0$, not $y = -5$.

Q4**B****B.** 30

Choice B is correct. Multiplying the left- and right-hand sides of the given equation by 25 yields $x^2 = 900$. Taking the square root of the left- and right-hand sides of this equation yields $x = 30$ or $x = -30$. Of these two solutions, only 30 is given as a choice.

Choice A is incorrect. This is a solution to the equation $x^2 = 36$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**B****B.** $d^2 + 11$

Choice B is correct. The given expression can be rewritten as $d^2 + 8 + 3$. Adding 8 and 3 in this expression yields $d^2 + 11$.

Choice A is incorrect. This expression is equivalent to $d^2 + 83$.

Choice C is incorrect. This expression is equivalent to $8 + d^2 - 3$.

Choice D is incorrect. This expression is equivalent to $-8 + d^2 - 3$.

Q6**A****A.** 4

Choice A is correct. It's given that $gx = x^2 + 9$. Substituting 25 for gx in this equation yields $25 = x^2 + 9$. Subtracting 9 from both sides of this equation yields $16 = x^2$. Taking the square root of each side of this equation yields $x = \pm 4$. It follows that $gx = 25$ when the value of x is 4 or -4. Only 4 is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**11**

Accept also: -7

The correct answer is 11 or -7. By the definition of absolute value, if $x - 2 = 9$, then $x - 2 = 9$ or $x - 2 = -9$. Adding 2 to both sides of the equation $x - 2 = 9$ yields $x = 11$. Adding 2 to both sides of the equation $x - 2 = -9$ yields $x = -7$. Thus, the given equation, $x - 2 = 9$, has two possible solutions, 11 and -7. Note that 11 and -7 are examples of ways to enter a correct answer.

Q8**A**

A. $x^9x + 5$

Choice A is correct. Since x is a factor of each term in the given expression, the expression is equivalent to $x^9x + x5$, or $x^9x + 5$.

Choice B is incorrect. This expression is equivalent to $45x^2 + 5x$, not $9x^2 + 5x$.

Choice C is incorrect. This expression is equivalent to $9x^2 + 45x$, not $9x^2 + 5x$.

Choice D is incorrect. This expression is equivalent to $9x^3 + 5x^2$, not $9x^2 + 5x$.

Q9**D**

D. 3

Choice D is correct. Because the graph of $y = f(x)$ is shown, the value of $f(0)$ is the value of y on the graph that corresponds with $x = 0$. When $x = 0$, the corresponding value of y is 3. Therefore, the value of $f(0)$ is 3.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q10**B**

B. 4

Choice B is correct. Subtracting 61 from each side of the given equation yields $p = 4$. By the definition of absolute value, if $p = 4$, then $p = 4$ or $p = -4$. Of the given choices, 4 is a solution to the given equation.

Choice A is incorrect. This is the quotient, not the difference, of 65 and 61.

Choice C is incorrect. This is the sum, not the difference, of 65 and 61.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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